Slide Set #8

Asymptotic Approximations Pt. I

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Notes

Basics

- **Sequences:** an ordered countably infinite collection of mathematical objects. A sequence is denoted using the generating rule, e.g.: $\{n^{-1}\}_{n\geq 1}$, or $\{n^{-1}\}_{n=1}^{\infty}$, often we simply write with write $\{n^{-1}\}$. The problem for real sequences is: what about the limit point (if one exists)?
- **Sequences of random variables:** the problem is to find out whether a limit probabilistic behavior exists
- **Sequence of estimates:** we imagine to repeat the random experiment infinitely many times. Under certain assumption one can study the limit behaviour of the sequence $\{\hat{\theta}_n\}_{n\geq 1}$ of estimates of θ
- **Stochastic process:** is a sequence of random scalars/vectors.

Notes

Consider a random variable X taking values in $\{0, 1\}$. Consider a iid sequence $\{X_n\}$, and suppose that the probability of X_n is given by

$$\Pr\{X_n = 0\} = f(n), \qquad \Pr\{X_n = 1\} = 1 - f(n)$$

For $n = 1, 2, \ldots,$ you have a sequence of sets of outcomes. Consider

$$E_n = \{\text{all the outcomes}: \ X_n = 0\}$$

Remark: as n changes, both E_n and $\Pr\{X_n\}$ keep changing.

In the limit (that is $n \to \infty$) we would like to known what happens to E_n, and what happes to its probability law

Notes

Assume also $f(n) < f(n-1)$. What's the limit behavior of the experiment?

- Intuitively one can think that as $n \to \infty$ the probability of E_n decreases monotonically ($\searrow$) so that in the limit E_n is "*rather unlikely*".
- This is not necessarily true because it depends what we mean by "*rather unlikely*".
- Different notions of "stochastic limit behavior" exists. We will see how the previous sequence converges in some specific sense.

Notes

Modes of convergence

- Convergence in probability
- Almost sure convergence (aka convergence with probability 1)
- Convergence in L_p norm. We cover L_2 which is mean square convergence (aka quadratic mean convergence)
- Convergence in distribution

Notes

Definition (Convergence in Probability)

A sequence of random scalars $\{z_n\}$ converges in probability to a constant α if for any $\varepsilon > 0$

$$\lim_{n\to\infty} \Pr(|z_n - \alpha| > \varepsilon) = 0 \tag{8.1}$$

The constant α is called the probability limit of z_n

Alternative notations: $\operatorname*{plim}_{n\to\infty} z_n = \alpha$, or $z_n \xrightarrow{\mathsf{p}} \alpha$

Think of this mode of convergence as a *convergence of sequences of real numbers*. In fact, denote

$$p_n := \Pr(\text{outcomes}: \ |z_n - \alpha| > \varepsilon),$$

then, (8.1) is equivalent to $\lim_{n\to\infty} p_n = 0$.

Notes

Definition (Almost Sure Convergence)

A sequence of random scalars $\{z_n\}$ converges almost surely to a constant α if

$$\Pr\left(\lim_{n\to\infty} z_n = \alpha\right) = 1 \tag{8.2}$$

Also denoted as $z_n \xrightarrow{\text{as}} \alpha$

Think of this convergence as "*set shrinkage*": consider the set

$$E_n := \{\text{outcomes: } |z_n - \alpha| \geq \varepsilon \text{ for all } n \geq N_\varepsilon\}$$

$\xrightarrow{\text{as}}$ means that for every $\varepsilon > 0$ there exists an N_ε such that for $n \geq N_\varepsilon$ the set E_n "shrinks to nothing" so that its probability is zero.

Hence, whatever small ε is picked, we always find e certain term of the sequence z_{N_ε} such that for all subsequent terms, the event that $|z_n - \alpha| \geq \varepsilon$ cannot happen anymore.

Notes

$\xrightarrow{\text{p}}$ vs $\xrightarrow{\text{as}}$

Story behind the example:

- a machine can fail at the nth trial ($X_n = 0$), or it keeps working ($X_n = 1$)
- the machine is like the best *Taurasi Wine* (check it out... in moderation!), it gets better as it gets older

$$\Pr\{\text{probability of failure}\} = \Pr\{X_n = 0\} \quad \text{decreases with } n$$

Intuitively, as we use the machine for an infinite number of trials, at the end $X = 1$ "*surely*". Any converge statement should agree with this. However converge statements mean different mechanisms:

- if $X_n \xrightarrow{\text{p}} 1$: probability of failure gets close to 0 after a sufficiently large number of usages
- if $X_n \xrightarrow{\text{as}} 1$: at some point you exhausted all failures (the machine works perfectly from some trial onward). That is, there exist an N such that for all $n \geq N$ all failures are exhausted

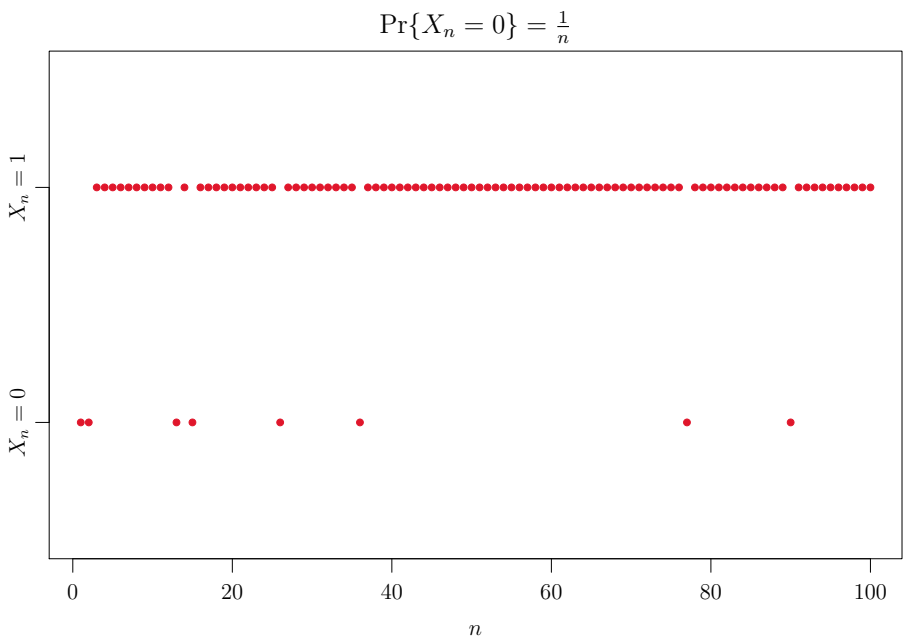
Notes

For $\xrightarrow{\text{as}}$ you know that there exists N such that for $n \geq N$ you stop having failures, but you don't know N. Therefore, it seems that in **practice** there is no difference between the two modes of convergence.

However, the difference is big! To see this we simulate two cases

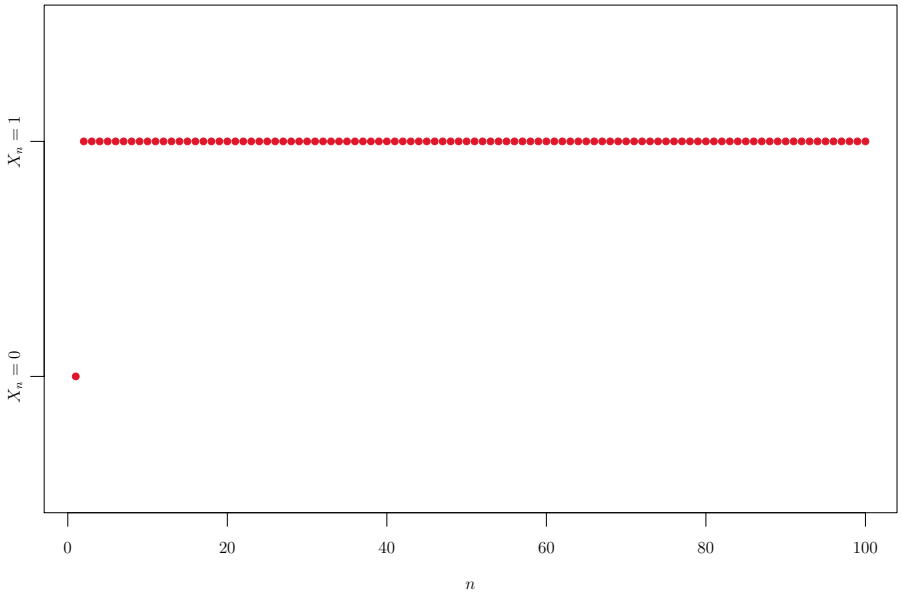
- Case 1: $\Pr\{X_n = 0\} = f(n) = \frac{1}{n}$. It can be shown that $X_n \xrightarrow{\text{p}} 1$ but almost sure convergence doesn't apply
- Case 2: $\Pr\{X_n = 0\} = f(n) = \frac{1}{n^2}$. It can be shown that both $X_n \xrightarrow{\text{as}} 1$ and $X_n \xrightarrow{\text{p}} 1$ apply

Notes



Notes

$\Pr\{X_n = 0\} = \frac{1}{n^2}$



Notes

Definition (Mean square convergence, or convergence in quadratic mean)

A sequence of random scalars $\{z_n\}$ with bounded second moment ($\mathrm{E}[z_n^2] < \infty,\ n = 1, 2, \dots$) converges in mean square to a constant α if

$$\lim_{n\to\infty} \mathrm{E}\left[(z_n - \alpha)^2\right] = 0 \tag{8.3}$$

This is often denoted as $z_n \xrightarrow{\text{ms}} \alpha$

As for the case of convergence in probability this is also connected to convergence of sequences of real numbers.

In fact denote $M_n := \mathrm{E}\left[(z_n - \alpha)^2\right]$, here we are looking at the limit point of $M_n \in \mathbb{R}$

Notes

Random vectors and matrices

these definitions of convergence (p, as, ms) can be extended to sequences of random vectors or random matrices by requiring *element-wise convergence*.

Random limits

In these definitions the limit is a constant. The limit can also be a random scalar z, or a random vector, $\boldsymbol{z}$.

The latter can be seen as follows: one can write $z_n \xrightarrow{\text{p}} z$ whenever $z_n - z \xrightarrow{\text{p}} 0$

Notes

Definition (Convergence in distribution, or weak convergence)

Assume

- Y is a random variable with cdf G
- $\{Z_n\}$ is a sequence of random variables, each with cdf F_n

$\{Z_n\}$ is said to converge in distribution to a random variable Y if

$$\lim_{n\to\infty} F_n(t) = G(t), \text{ for all points } t \text{ at which } G \text{ is continuous.}$$

The latter is denoted with $Z_n \xrightarrow{\text{d}} Y$

Why "*Weak*"

- Requires only point-wise convergence of functions: we just have to check that $\lim_{n\to\infty} |F_n(t) - G(t)| = 0$ for all points t where $G(t)$ is continuous
- Doesn't relate outcomes. Here we do not compare outcomes from Z_n vs Y, we only compare their cdf.

Notes

Extension to sequences of random vectors

$z_n \xrightarrow{\text{d}} z$ if the joint cdf of z_n converges (point-wise) to the joint cdf of the elements of z at every continuity point

⚠ Element-wise convergence in distribution for random vectors (i.e. convergence in distribution for the marginals of z_n) does not necessarily imply convergence for the whole vector z_n

Observe: the inner product of 2 vectors is a real valued function

$$\boldsymbol{\lambda}'\boldsymbol{z} \;=\; \sum_{k=1}^{K} \lambda_k z_k \;\in\; \mathbb{R}$$

Theorem (Cramér-Wold, 1936)

Let $\{z_n\}$ be a sequence of K-dimensional random vectors. Then for every $\boldsymbol{\lambda} \in \mathbb{R}^K$

$$z_n \xrightarrow{\text{d}} z \quad \Longleftrightarrow \quad \boldsymbol{\lambda}' z_n \xrightarrow{\text{d}} \boldsymbol{\lambda}' z$$

Notes

Remarks on the convergence of moments

Theorem (Weak convergence vs sequences of moments)

Take $r \geq 1$, consider the sequence $\{z_n\}$, and assume

- $z_n \xrightarrow{\text{d}} z$
- $\sup_n |z_n^r| \leq y$ with $\mathrm{E}[y] < \infty$ (*)

Then

- $\mathrm{E}[|z^r|] < +\infty$
- $\lim_{n\to\infty} E[z_n^r] = \mathrm{E}[z^r]$

Remark: (*) is a sufficient condition for the "*uniform integrability*" of z_n^r. Uniform integrability allows to to interchange integration and limits.

Notes

Relationships among modes of convergence

- $\boldsymbol{z}_n \xrightarrow{\text{ms}} \boldsymbol{z} \implies \boldsymbol{z}_n \xrightarrow{\text{p}} \boldsymbol{z}$
- $\boldsymbol{z}_n \xrightarrow{\text{as}} \boldsymbol{z} \implies \boldsymbol{z}_n \xrightarrow{\text{p}} \boldsymbol{z}$
- $\boldsymbol{z}_n \xrightarrow{\text{p}} \boldsymbol{z} \implies \boldsymbol{z}_n \xrightarrow{\text{d}} \boldsymbol{z}$

Under certain conditions (uniform integrability, and/or convergence rates):

- convergence in probability $\implies$ mean square convergence
- mean square convergence $\implies$ almost sure convergence

Notes

Key tools

Theorem (Continuous mapping theorem)

$\{\boldsymbol{z}_n\}$ is a sequence of random vectors, $\boldsymbol{g}(\cdot)$ is a vector-valued continuous function. Then (assuming existence of the following limits)

1. $\boldsymbol{z}_n \xrightarrow{\text{as}} \boldsymbol{z} \implies \boldsymbol{g}(\boldsymbol{z}_n) \xrightarrow{\text{as}} \boldsymbol{g}(\boldsymbol{z})$
2. $\boldsymbol{z}_n \xrightarrow{\text{p}} \boldsymbol{z} \implies \boldsymbol{g}(\boldsymbol{z}_n) \xrightarrow{\text{p}} \boldsymbol{g}(\boldsymbol{z})$
3. $\boldsymbol{z}_n \xrightarrow{\text{d}} \boldsymbol{z} \implies \boldsymbol{g}(\boldsymbol{z}_n) \xrightarrow{\text{d}} \boldsymbol{g}(\boldsymbol{z})$

An immediate implication is that most operations preserve convergence:

- $x_n \xrightarrow{\text{p}} \beta$, $y_n \xrightarrow{\text{p}} \gamma$, this implies that $x_n + y_n \xrightarrow{\text{p}} \beta + \gamma$, $x_n y_n \xrightarrow{\text{p}} \beta\gamma$, $x_n / y_n \xrightarrow{\text{p}} \beta/\gamma$
- also useful for matrices: $\boldsymbol{Y}_n \xrightarrow{\text{p}} \boldsymbol{\Gamma}$ implies that $\boldsymbol{Y}_n^{-1} \xrightarrow{\text{p}} \boldsymbol{\Gamma}^{-1}$, provided that these are invertible.

Notes

Theorem (Slutzky, 1925)

Let $\{\boldsymbol{x}_n\}$ and $\{\boldsymbol{y}_n\}$ be sequences of random vectors. Assume the following limit exist

- $\boldsymbol{x}_n \xrightarrow{\text{d}} \boldsymbol{x}$, where $\boldsymbol{x}$ is a (vector) random variable
- $\boldsymbol{y}_n \xrightarrow{\text{p}} \boldsymbol{\alpha}$, where $\boldsymbol{\alpha}$ is a constant (vector)

Then

1. $\boldsymbol{x}_n + \boldsymbol{y}_n \xrightarrow{\text{d}} \boldsymbol{\alpha} + \boldsymbol{x}$
2. $\boldsymbol{x}_n \boldsymbol{y}_n \xrightarrow{\text{d}} \boldsymbol{\alpha}\boldsymbol{x}$

"*Slutzky's Lemma*" is central because

- mixes two different notions of convergence
- says that addition(differences) and product(division) between a sequence converging in distribution, and a sequence converging in probability, ultimately converge in distribution!

Notes

Continuous Mapping + Slutzky Theorem = *Machine Gun*

Linear combinations

using Slutzky's assumptions we can prove that in general

- $\boldsymbol{y}_n' \boldsymbol{x}_n \xrightarrow{\text{d}} \boldsymbol{\alpha}' \boldsymbol{x}$
- $g(\boldsymbol{y}_n' \boldsymbol{x}_n) \xrightarrow{\text{d}} g(\boldsymbol{\alpha}' \boldsymbol{x})$ for any continuous map g

Linear and quadratic maps

assume that $\{\boldsymbol{A}_n\}$ is a sequence of random matrices, $\boldsymbol{A}$ is an appropriate matrix, suppose $\boldsymbol{A}_n \xrightarrow{\text{p}} \boldsymbol{A}$ and $\boldsymbol{x}_n \xrightarrow{\text{d}} \boldsymbol{x}$, then

- $\boldsymbol{A}_n \boldsymbol{x}_n \xrightarrow{\text{d}} \boldsymbol{A}\boldsymbol{x}$.
- $\boldsymbol{x}_n' \boldsymbol{A}_n \boldsymbol{x}_n \xrightarrow{\text{d}} \boldsymbol{x}' \boldsymbol{A} \boldsymbol{x}$

Notes

Asymptotic equivalence

Suppose we need to find the limit behavior of $\boldsymbol{z}_n$, but this is too difficult

In certain cases we are able to decompose $\boldsymbol{z}_n = \boldsymbol{x}_n + \boldsymbol{y}_n$, where

- $\{\boldsymbol{x}_n\}$ and $\{\boldsymbol{y}_n\}$ are both *simpler* to study than $\{\boldsymbol{z}_n\}$
- $\boldsymbol{y}_n \xrightarrow{\text{p}} \boldsymbol{0}$

so that $\boldsymbol{z}_n - \boldsymbol{x}_n \xrightarrow{\text{p}} \boldsymbol{0}$. This means that asymptotically $\boldsymbol{z}_n$ behaves like $\boldsymbol{x}_n$, e.g. they are *asymptotically equivalent* and we write $\boldsymbol{z}_n \overset{\text{a}}{\approx} \boldsymbol{x}_n$.

Why it is useful to find an *asymptotic equivalent*? Because sometimes we can characterize an asymptotic behavior looking at something equivalent but simpler to manipulate

Note: another notation for asymptotic equivalence is $\boldsymbol{z}_n = \boldsymbol{x}_n + o_p$, where $o_p =$"random sequence converging to $\boldsymbol{0}$ in probability"

Notes

Theorem (Delta method; Doob, 1935)

Let $\{\boldsymbol{x}_n\}$ be a sequence of K-dimensional random vectors. Assume

- $\boldsymbol{x}_n \xrightarrow{\text{p}} \boldsymbol{\beta}$, where $\boldsymbol{\beta} \in \mathbb{R}^K$ is non-random
- $\sqrt{n}\{\boldsymbol{x}_n - \boldsymbol{\beta}\} \xrightarrow{\text{d}} \boldsymbol{z}$
- $\boldsymbol{t} \in \mathbb{R}^K$, and $\boldsymbol{g}(\boldsymbol{t})$ is a vector-valued function $\boldsymbol{g} : \mathbb{R}^K \to \mathbb{R}^J$, with continuous 1st -order derivatives
- $\boldsymbol{A}(\boldsymbol{\beta})$ is the the Jacobian matrix of $\boldsymbol{g}(\boldsymbol{t})$ evaluated at $\boldsymbol{t} = \boldsymbol{\beta}$.

Then

$$\sqrt{n}\{\boldsymbol{g}(\boldsymbol{x}_n) - \boldsymbol{g}(\boldsymbol{\beta})\} \xrightarrow{\text{d}} \boldsymbol{A}(\boldsymbol{\beta})\boldsymbol{z}$$

In most cases we work with normal limits like

$$\sqrt{n}\{\boldsymbol{x}_n - \boldsymbol{\mu}\} \xrightarrow{\text{d}} \mathcal{N}(\boldsymbol{0}, \boldsymbol{\Sigma})$$

The delta method theorem is very useful to derive the distribution of functions of sequences with a normal limit

Notes

Example

Suppose $\{\boldsymbol{\beta}_n\}$ is a random sequence, and $\boldsymbol{\beta}_0$ is a fixed vector in $\mathbb{R}^2$

$$\boldsymbol{\beta}_n = \begin{pmatrix} \beta_{1n} \\ \beta_{2n} \end{pmatrix} \quad \boldsymbol{\beta}_0 = \begin{pmatrix} 3 \\ 7 \end{pmatrix}$$

Suppose also that, for $\sigma^2 = 4$

$$\sqrt{n}\,\{\boldsymbol{\beta}_n - \boldsymbol{\beta}_0\} \xrightarrow{\mathsf{d}} \mathcal{N}(\mathbf{0}, \sigma^2 \boldsymbol{I}_2)$$

Take $\boldsymbol{t} = (t_1, t_2)'$, and the vector-valued function $\boldsymbol{y} = \boldsymbol{g}(\boldsymbol{t})$ where

$$\boldsymbol{y} = \boldsymbol{g}(\boldsymbol{t}) = \begin{cases} y_1 = & t_1 + 2t_2 \\ y_1 = & t_1^2 + t_2^3 \end{cases}$$

Notes

Let $\boldsymbol{y}_0 = \boldsymbol{g}(\boldsymbol{\beta}_0)$, replacing $\boldsymbol{\beta}_0$ into $\boldsymbol{g}(\cdot)$ we get

$$\boldsymbol{y}_0 = \begin{pmatrix} 17 \\ 352 \end{pmatrix}$$

The Jacobian matrix of $\boldsymbol{g}(\cdot)$ is

$$\boldsymbol{A}(\boldsymbol{t}) = \begin{pmatrix} 1 & 2 \\ 2t_1 & 3t_2^2 \end{pmatrix}$$

therefore for $\boldsymbol{t} = \boldsymbol{\beta}_0 = (3, 7)'$ we have

$$\boldsymbol{A}(\boldsymbol{\beta}_0) = \begin{pmatrix} 1 & 2 \\ 6 & 147 \end{pmatrix}$$

Notes

Application of the Delta Method Theorem tell us that

$$\sqrt{n}\left\{\boldsymbol{g}(\boldsymbol{\beta}_n) - \boldsymbol{g}(\boldsymbol{\beta}_0)\right\} \xrightarrow{\text{d}} \boldsymbol{A}(\boldsymbol{\beta}_0)\,\mathcal{N}(\boldsymbol{0}, \sigma^2 \boldsymbol{I}_2)$$

The var matrix of the rhs is given by

$$\boldsymbol{V} = \boldsymbol{A}(\boldsymbol{\beta}_0)\sigma^2 \boldsymbol{I}_2 \boldsymbol{A}(\boldsymbol{\beta}_0)' = \sigma^2 \boldsymbol{A}(\boldsymbol{\beta}_0)\boldsymbol{A}(\boldsymbol{\beta}_0)' = \begin{pmatrix} 50 & 1200 \\ 1200 & 86500 \end{pmatrix}$$

Therefore,

$$\sqrt{n}\left\{\boldsymbol{g}(\boldsymbol{\beta}_n) - \boldsymbol{y}_0\right\} \xrightarrow{\text{d}} \mathcal{N}(\boldsymbol{0}, \boldsymbol{V})$$

Notes

Laws of large numbers (LLN)

Consider the sequence $\{z_i\}$. Define the arithmetic mean of random variables

$$\bar{z}_n = \frac{1}{n}\sum_{i=1}^{n} z_i$$

LLNs study the asymptotic limit of the new sequence $\{\bar{z}_n\}$

Theorem (Chebychev's weak LLN)

Suppose $\lim_{n\to\infty} \mathrm{E}[\bar{z}_n] = \mu$ and $\lim_{n\to\infty} \mathrm{Var}[\bar{z}_n] = 0$, then $\bar{z}_n \xrightarrow{\text{p}} \mu$

Theorem (Kolmogorov's strong LLN)

Let $\{z_i\}$ be i.i.d. with $\mathrm{E}[z_i] = \mu < \infty$, then $\bar{z}_n \xrightarrow{\text{as}} \mu$.

These LLNs can be easily extended to random vectors by requiring element-wise convergence

Notes

Central limit theorems (CLT)

CLTs are weak convergence results for scaled centered sequences

$$c_n(\bar{z}_n \; - \; \mathrm{E}[\bar{z}_n])$$

where c_n is an appropriate non-random scaling sequence

Theorem (Lindeberg-Levy, 1920)

Let $\{\boldsymbol{z}_i\}$ be i.i.d. with $\mathrm{E}[\boldsymbol{z}_i] = \boldsymbol{\mu}$ and $\mathrm{Var}[\boldsymbol{z}_i] = \boldsymbol{\Sigma}$ for all $i = 1, 2, \cdots, n$, then

$$\sqrt{n}(\overline{\boldsymbol{z}}_n - \boldsymbol{\mu}) \; = \; \frac{1}{\sqrt{n}} \sum_{i=1}^{n} (\boldsymbol{z}_i - \boldsymbol{\mu}) \; \xrightarrow{\mathsf{d}} \; \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma})$$

Speed of convergence and scaling sequence

Here $c_n = \sqrt{(n)}$. By the LLN, $\bar{\boldsymbol{z}}_n - \boldsymbol{\mu} \xrightarrow{\mathsf{p}} \mathbf{0}$. Therefore, asymptotically the distribution of $\{\bar{\boldsymbol{z}}_n - \boldsymbol{\mu}\}$ is degenerate at $\mathbf{0}$. In order to have a proper limit distribution for $\{\bar{z}_n - \mu\}$, one needs to *inflate* it by a factor which grows at the same speed at which $\{\bar{\boldsymbol{z}}_n - \boldsymbol{\mu}\}$ vanishes. In most cases such factor is $c_n = \sqrt{n}$.

Notes

Notes